Dynamic Causal Character Graphs: Structured State Control for Persistent Fictional Agents

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Abstract

Large language models (LLMs) struggle to maintain coherent character personas in long-form interactions due to their autoregressive nature, which lacks an explicit model of persistent state. This leads to issues like personality drift and inconsistent motivations. We introduce Causal Character Graphs (CCGs), a lightweight, structured state representation that models a character's core traits, beliefs, and world constraints as a directed acyclic graph. Our central hypothesis is that persistent structured state, not post-hoc critique, is the primary driver of long-horizon character coherence. Planning and verification act as auxiliary stabilizers, but coherent behavior emerges from causal state propagation itself.

1 Introduction

Large language models generate dialogue through autoregressive next-token prediction, a mechanism that lacks an explicit representation of persistent psychological or world-level structure. While effective for short exchanges, this results in personality drift, inconsistent motivations, temporal leakage, and violations of fictional-world constraints in long-form interactions. Core traits, beliefs, and narrative boundaries are implicitly encoded in context tokens and are therefore susceptible to overwriting as conversations progress. We believe that this instability stems from the absence of a structured latent state governing trait persistence, belief revision, emotional transitions, and temporal grounding. To address this, we introduce Dynamic Causal Character Graphs (DCCG), a lightweight causal framework that models a character's core traits, beliefs, and world constraints. Dialogue generation is decomposed into two phases: (1) an Update phase that performs structured state transitions via causal propagation, and (2) an Act phase that conditions generation on the updated state. Unlike memory buffers or summarization-based approaches, DCCG enforces explicit causal dependencies and temporally consistent state evolution, enabling controlled character evolution and sustained narrative coherence.

2 Problem Definitions

Current character-based dialogue systems exhibit recurring failure modes in long-horizon interactions (Araujo et al., 2026), including personality drift, motivational inconsistency, temporal leakage, fictional-world violations, and unstructured character evolution. Additionally, existing systems frequently conflate objective world facts with a character's epistemic knowledge, resulting in temporal leakage and spoiler-like behavior (Park et al., 2023). These failures arise because character state is not maintained as an explicit, persistent representation; instead, it is implicitly encoded in the transient context window, which functions as working memory and is subject to truncation and drift over long interactions (Zhao et al., 2026). As conversations progress, core traits, goals, and narrative constraints are gradually overwritten by recent dialogue. Existing approaches such as longer context windows, summarization buffers, memory retrieval, and fine-tuning improve recall but do not enforce structured state transitions. In particular, they lack an explicit causal model linking traits, beliefs, and actions, leaving character behavior unstable over extended interactions.

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3 Methods

We aim to introduce Dynamic Causal Character Graphs (DCCGs), a structured framework for controlling long-horizon character behavior in dialogue systems. The core idea is to replace implicit context-based personality encoding with an explicit, lightweight causal state representation that guides belief updates and action generation.

Unlike dependence graphs or memory buffers, DCCG uses an SCM-inspired state transition layer with explicit mechanisms and intervention semantics. We do **not** claim causal identification from observational dialogue logs; rather, we use SCM structure as an engineering prior for modular state updates, explicit intervention control, and counterfactual consistency checks. In an industry setting, this framing provides auditable state transitions and controllable failure analysis while remaining compatible with frozen foundation models. External events are treated as exogenous inputs to these mechanisms, enabling principled belief revision and constrained propagation of emotion and behavioral responses.

At each dialogue turn, generation is decomposed into two phases: a structured state update phase and a conditioned action generation phase. The internal character state is represented as a time-unrolled dynamic causal graph (Dynamic SCM) whose nodes encode psychological variables and world constraints, and whose edges define causal influence between variables.

3.1 Character State Representation

Each character is defined by a directed causal graph

$$G = (V, E)$$

where nodes represent structured components of the character's internal state and edges define causal dependencies between components.

3.1.1 State Schema

The node set V is organized into a compact set of psychologically interpretable state variables. The graph is intentionally kept small (typically tens of nodes) to ensure stable long-horizon behavior while allowing rich belief representations.

- **Traits and Values**
  Stable personality characteristics and value orientations are represented as scalar variables

  $$\tau_j \in [-1, 1] \text{ or } [0, 1]$$

  Examples include bravery, honesty, curiosity, and risk-aversion. These nodes have low plasticity and evolve slowly over time.
- **Emotional State**
  Emotion is represented using a low-dimensional affective state:

  $$\mathbf{e}_t = (v_t, a_t)$$

  where v_t denotes valence and a_t denotes arousal. Discrete emotional tags (e.g., fear, anger, joy) are represented as intensity values in $[0, 1]$ and provide interpretable labels for dialogue generation.
- **Relationships**
  For each entity x, relationship state is represented by scalar variables:

  $$R_t(x) = (\text{trust}, \text{affection}, \text{respect}) \in [0, 1]^3$$

  These nodes evolve based on interaction history and influence belief revision and action selection.

- **Beliefs**
  Beliefs are represented as a set of propositions

  $$B_t^c = \{b_k\}$$

  where each belief

  $$b_k = (p_k, S_k)$$

  consists of:

  - confidence probability $p_k \in [0, 1]$
  - supporting evidence identifiers S_k

  This representation allows uncertainty, false beliefs, and provenance tracking.
- **Intentions**
  Intentions represent the character's current behavioral goals and dialogue plans. These nodes typically have high plasticity and depend on beliefs and emotional state.
- **Timeline State**
  Temporal grounding is maintained by a timeline index

  $$T_t \in \mathbb{Z}$$

  together with a knowledge boundary indicating the maximum time index accessible to the character.
- **World Constraints**
  Explicit world constraints encode immutable narrative rules such as:

  - forbidden abilities
  - physical limitations
  - knowledge restrictions

  These constraints act as hard conditions during dialogue generation.

3.1.2 Causal Dependencies

Edges E define directed causal dependencies between state variables. A canonical dependency structure is:

$$\text{Traits} \rightarrow \text{Beliefs} \rightarrow \text{Emotions} \rightarrow \text{Intentions} \rightarrow \text{Dialogue}$$

while allowing feedback across time steps through the dynamic causal model.

3.1.3 Structural Update Model

Each node $v_i \in V$ maintains a structured internal state

$$s_{t,i}$$

together with a plasticity parameter

$$\alpha_i \in [0, 1]$$

that governs the rate of state evolution.

We model the character graph as a structural causal system. At dialogue turn t, node i is updated via a structural mechanism followed by plasticity-controlled integration:

$$\tilde{s}_{t,i} = f_i(\mathrm{Pa}_i(\mathbf{s}_{t-1}), e_t) \tag{1}$$
$$s_{t,i} = (1-\alpha_i)s_{t-1,i} + \alpha_i \tilde{s}_{t,i} \tag{2}$$

where:

- $\mathbf{s}_{t-1}$ denotes the vector of node states at time $t-1$
- $\mathrm{Pa}_i(\cdot)$ returns the parent nodes of v_i
- e_t is an exogenous intervention or event
- f_i is the structural mechanism governing node i
- $\tilde{s}_{t,i}$ is the proposed update before plasticity filtering
- α_i is the base plasticity parameter

3.1.4 Interventions and $do(\cdot)$ Semantics

To make the causal claims operational, we define DCCG as a controlled transition system with SCM-style interventions.

Let $\mathbf{s}_t = \{s_{t,i}\}_{i=1}^{|V|}$ denote endogenous character-state variables and let $\mathbf{u}_t$ denote exogenous inputs (user utterance events, narrator actions, or simulator events). For each node i we specify a local mechanism

$$s_{t,i} = F_i\big(\mathrm{Pa}_i(\mathbf{s}_{t-1}), \mathbf{u}_t\big)$$

followed by plasticity-controlled integration as defined above. Edges in E are therefore not only an execution order; they specify which parent set is allowed to influence each mechanism F_i.

An intervention on node j at turn t is represented as:

$$do\big(s_{t,j} = \bar{s}\big)$$

which replaces the native mechanism F_j for that turn and clamps $s_{t,j}$ to $\bar{s}$ before downstream updates are computed. This supports two practically important intervention types:

- **World interventions:** direct edits to objective world variables in W_t (e.g., a new global event, location change, or constraint toggle).
- **Character interventions:** direct edits to belief, emotion, intention, or relationship nodes for controlled simulation and debugging.

This formulation is intentionally *interventional* rather than *identificational*: we use known structural assignments to control and audit behavior, not to estimate causal effects from passive text-only observations.

The SCM framing adds value in three concrete ways: (1) **counterfactual consistency checks** by comparing generated responses under alternate $do(\cdot)$ assignments, (2) **intervention-based debugging** by localizing failures to specific mechanisms F_i, and (3) **epistemic/world disentanglement** by separating interventions on world truth (W_t) from updates to subjective beliefs (B_t^c).

Operational Intervention Protocol For deployment and evaluation, each turn executes the following protocol:

1. Apply any authorized intervention set
$$\mathcal{D}_t = \{do(s_{t,j} = \bar{s}_j)\}_{j \in J_t}$$
to world or character nodes.
2. Update all non-intervened nodes using their native mechanisms F_i and parent values.

3. Run downstream consistency checks (belief, temporal, and constraint checks) before response emission.
4. Log intervention metadata and violated checks for post-hoc debugging and policy tuning.

This makes the notion of "causal control" concrete: interventions modify specific structural assignments, and only descendants can change as a consequence.

Counterfactual Query Interface Given the same prior state and input, we define a counterfactual comparison as:

$$\Delta y_t = \mathrm{Compare}\left(P(y_t \mid \mathbf{s}_{t-1}, u_t), P\big(y_t \mid do(s_{t,j} = \bar{s}), \mathbf{s}_{t-1}, u_t\big)\right)$$

where $\mathrm{Compare}(\cdot)$ measures differences in claims, tone, or policy compliance. This interface supports practical diagnostics such as "Would the response remain safe and in-character if trust were lower?"

3.1.5 Plasticity Regimes

Nodes are grouped by plasticity:

- Stable nodes (traits and values) have small α
- Semi-stable nodes (beliefs and relationships) have moderate α
- Fast nodes (emotions and intentions) have large α

This structure ensures gradual evolution without abrupt personality drift while still allowing meaningful character development.

3.2 Two-Phase Dialogue Loop

Each interaction is divided into two steps: an **Update Phase** (internal processing) followed by an **Act Phase** (dialogue generation). This separation ensures that the character's internal state is coherent before producing a response.

3.2.1 Phase 1: Update Phase

Before the character responds, the system incorporates the user's input and story context to update the internal state.

1. **Event Extraction**
   The user's message u_t is converted into a structured event representation

   $$e_t$$

   using a schema-constrained language model extraction pass followed by a validation step. The extracted event frame has the form

   $$e_t = (\mathcal{P}_t, \mathcal{E}_t, a_t, \tau_t, c_t)$$

   where:

   - $\mathcal{P}_t$ is the set of asserted propositions
   - $\mathcal{E}_t$ is the set of referenced entities
   - a_t is the speaker identity
   - τ_t is the inferred emotional tone
   - $c_t \in [0, 1]$ is the extraction confidence score

The extraction model is constrained to produce structured outputs conforming to a predefined schema to ensure consistent state updates.

To reduce silent extraction errors, a second language model pass acts as a validation mechanism. The validator evaluates whether the extracted event frame is consistent with the input message and satisfies schema constraints. Invalid or contradictory event frames are rejected and re-extracted.

The validation model also produces an extraction confidence estimate c_t, which is used to control the strength of subsequent belief updates.

2. **Belief Revision Policy + Epistemic State**

3.3 Epistemic State and Belief Revision

We define the character's belief system as a dynamic probabilistic graph that evolves via recursive evidence accumulation. This approach replaces heuristic updates with a stable log-odds framework (**?**), ensuring numerical stability and proper handling of conflicting evidence.

3.3.1 Probabilistic Belief Representation

The Character Belief Graph B_t^c is a set of weighted propositions. Each belief node $b_k \in B_t^c$ is defined as:

$$b_k = (\phi_k, \ell_{t,k}, \mathcal{S}_k)$$

where:

- ϕ_k: The propositional content (e.g., `Player_has_Item(Key)`).
- $\ell_{t,k} \in \mathbb{R}$: The epistemic confidence expressed as log-odds (logit).
- $\mathcal{S}_k$: Provenance metadata tracking the source of the belief.

We utilize log-odds representation to allow for additive evidence updates:

$$\ell_{t,k} = \ln\left(\frac{p_{t,k}}{1 - p_{t,k}}\right) \tag{3}$$

An accumulated log-odds of $\ell = 0$ represents maximum uncertainty ($p = 0.5$).

3.3.2 Recursive Evidence Integration

At time t, an extracted event e_t acts as an observation. We model the belief update rule as:

$$\ell_{t,k} = \ell_{t-1,k} + \underbrace{\lambda_{base} \cdot \sigma(e_t)}_{\text{Shock-Gated Plasticity}} \cdot \rho(src) \cdot \delta(e_t, \phi_k) \cdot c_t \tag{4}$$

where the components are defined as:

Shock-Gated Plasticity $\lambda_{base} \cdot \sigma(e_t)$ To distinguish justified evolution from random drift, we modulate the base learning rate λ_{base} by a "Narrative Importance" factor $\sigma(e_t)$, defined as a function of of event surprisal/narrative importance. Routine dialogue has low importance ($\sigma \approx 1$), preventing drift, while high-surprisal events (schema violations) yield high importance ($\sigma \gg 1$), allowing for rapid, justified character development.

Operational Definition of Narrative Importance We compute event surprisal as:

$$\text{surprisal}(e_t) = -\log P(e_t \mid h_{t-1})$$

where h_{t-1} denotes the prior dialogue history and $P(\cdot)$ is estimated using a frozen background language model or lightweight narrative prior.

Narrative importance is then defined as:

$$\sigma(e_t) = 1 + \beta \cdot \text{Normalize}\big(\text{surprisal}(e_t)\big)$$

where β controls shock sensitivity and Normalize$(\cdot)$ rescales surprisal to a bounded range.

This formulation ensures that only statistically unexpected or schema-violating events induce large effective plasticity, while routine dialogue yields $\sigma(e_t) \approx 1$.

Directional Alignment $\delta(e_t, \phi_k) \in \{-1, 0, 1\}$ Directional alignment is implemented as a deterministic predicate comparison layer over structured event frames and belief predicates. Specifically:

- If the event frame explicitly asserts ϕ_k, then $\delta = +1$
- If it asserts $\neg\phi_k$, then $\delta = -1$
- Otherwise $\delta = 0$

This comparison is performed prior to belief update and does not involve free-form language model reasoning, ensuring that belief revision is auditable and not implicitly aligned to generated text.

Source Credibility $\rho(src) \in [0, 1]$ This term weighs the update based on the character's trust in the source, derived directly from Relationship nodes R_t (Sperber et al., 2010):

$$\rho(src) = \begin{cases} 1.0 & \text{if } src = \text{DirectObservation} \\ R_t(src).\text{trust} & \text{if } src = \text{Character/User} \end{cases}$$

Evidence Strength $c_t \in [0, 1]$ The extraction confidence score provided by the validation model.

3.3.3 Causal Integration and Interventions

Belief nodes b_k serve as parent nodes in the larger Structural Causal Model (SCM), propagating influence to downstream "Emotion" and "Intention" nodes. This structure enables precise causal interventions:

$$do(b_k = \bar{x})$$

Operators can inject counterfactual beliefs (e.g., forcing a character to incorrectly believe a door is unlocked) to validate behavioral integrity without modifying the underlying narrative history.

Formally, the belief update rule defined above constitutes the structural mechanism F_i for epistemic nodes. These updates propagate through the graph edges E to influence downstream variables (e.g., triggering an emotional response to a realization) according to the general DCCG transition dynamics.

3.3.4 Consistency and Scalability

Computational Complexity Unlike full Bayesian Network inference which is NP-hard, the proposed additive log-odds update is an $O(1)$ scalar operation per active belief. This ensures the system remains scalable to simulations with hundreds of concurrent characters ($N > 100$) and millions of tokens without incurring the quadratic attention costs of full-context re-processing.

Conflict Resolution When two beliefs b_i and b_j represent mutually exclusive states (e.g., location contradictions), we apply pairwise normalization to maintain consistency (with ε smoothing to avoid dividing by 0):

$$p'_i = \frac{p_i}{p_i + p_j}, \quad p'_j = \frac{p_j}{p_i + p_j}$$

Closed-Loop Safeguard To prevent "hallucination loops," self-generated dialogue is **not** admissible as evidence. The source function $\rho(src)$ forces an update of 0 if the source is the character's own previous output. Only external observations (O_t) and validated user inputs constitute evidence.

3.3.5 Phase 2: Act Phase

The dialogue response y_t is generated by conditioning the language model on the updated character state and permissible context:

$$P(y_t \mid \mathbf{s}_t,\ h_t,\ u_t) \tag{5}$$

where:

- $\mathbf{s}_t$ denotes the updated character state
- h_t denotes the story context
- u_t denotes the user input

By separating state transition from language generation, the model enforces personality persistence, intervention-consistent coherence, and temporal consistency across long-horizon interactions.

We define the character system as a **Dynamic Structural Causal Model (D-SCM)**. At time step t, the system maintains two disjoint graph states.

3.3.6 Dual-Graph State Representation

To prevent temporal leakage and enforce epistemic consistency, we separate objective world state from character belief state.

At time step t, the system maintains two distinct state representations:

- **World Graph** (W_t)
  The world graph encodes objective narrative facts and timeline state. This graph represents canonical story truth and evolves according to external events and author-defined rules.
  Example variables include:
  - entity locations
  - object states
  - global events
  - timeline progression
- **Character Belief Graph** (B_t^c)
  We represent the character's belief system as a dynamic probabilistic graphical model that evolves via recursive Bayesian accumulation. This approach addresses the "ambiguity" of heuristic updates by grounding belief revision in log-odds evidence integration (Jaynes 2003), ensuring numerical stability and proper handling of conflicting evidence. The belief graph represents the subjective epistemic state of character c. This graph contains uncertain and potentially incorrect beliefs:

  $$B_t^c = \{b_k\}$$

  where each belief

  $$b_k = (p_k, S_k)$$

  consists of:
  - confidence probability $p_k \in [0, 1]$
  - supporting evidence identifiers S_k

  Unlike W_t, the belief graph may contain incomplete or incorrect information.

Observation Model Characters never condition directly on the world graph. Instead, belief updates are driven by observations derived from the world state:

$$O_t = \text{Perception}(W_t, \text{State}_c)$$

where State_c includes character location, attention, and sensory access.

Observations are converted into event representations e_t that drive structural updates in the character graph.

This separation ensures that dialogue generation depends only on the character's epistemic state rather than global narrative truth, preventing spoiler leakage and enforcing knowledge constraints.

3.3.7 Dynamic Structural Updates

The character state evolves as a time-unrolled dynamic structural causal model.
Each node v_i maintains state

$$s_{t,i}$$

updated according to the structural mechanism

$$\tilde{s}_{t,i} = f_i(\mathrm{Pa}_i(\mathbf{s}_{t-1}), e_t) \tag{6}$$
$$s_{t,i} = (1 - \alpha_{t,i}) s_{t-1,i} + \alpha_{t,i} \tilde{s}_{t,i} \tag{7}$$

where $\alpha_{t,i}$ is an effective plasticity parameter.

Shock-Gated Plasticity Plasticity is modulated by event characteristics to allow rare but meaningful character change:

$$\alpha_{t,i} = \alpha_i \, g(e_t)$$

where

$$g(e_t) = g(\text{surprisal}(e_t), \text{intensity}(e_t), \text{repetition}, \text{trust})$$

High-intensity or surprising events produce larger effective plasticity, allowing semi-stable variables such as beliefs and relationships to shift rapidly while stable traits remain resistant to change.
This mechanism enables realistic narrative arcs while preventing gradual personality drift.

3.3.8 Plan–Generate–Verify Act Phase

Dialogue generation is performed through a structured three-stage process.

Planning An intermediate dialogue plan is computed from the updated state:

$$\mathcal{I}_t = \text{Plan}(B_t^c, \mathbf{s}_t)$$

The plan specifies communicative intent and high-level response strategy (e.g., informing, deflecting, persuading).
Introducing a planning node provides a controllable bridge between structured state and surface language generation.

Generation A candidate response is generated by conditioning on the belief graph and dialogue plan:

$$\hat{y}_t \sim P(y_t \mid B_t^c, \mathcal{I}_t, u_t)$$

Verification (Local Consistency Checks Only) The verifier operates as a constraint checker rather than a full semantic controller. It receives:

- The candidate response $\hat{y}_t$
- A read-only snapshot of high-confidence beliefs
- The character's knowledge boundary
- Explicit world constraints

The verifier does **not** have access to:

- Structural mechanisms F_i
- Plasticity parameters
- Full causal parent structure
- Alternative counterfactual states

Its role is restricted to detecting violations, not rewriting behavior according to new objectives. If violations are found, a targeted revision prompt is issued that explicitly references the violated constraint.

Verification checks include:

- **Belief Consistency**
  Does $\hat{y}_t$ contradict high-confidence beliefs?
- **Temporal Consistency**
  Does $\hat{y}_t$ reference events beyond the character's knowledge boundary?
- **Constraint Consistency**
  Does $\hat{y}_t$ violate world constraints or character abilities?

If violations are detected, the response is revised using a targeted critique prompt.

This verification loop ensures that locally likely language model outputs remain globally consistent with the character state.

Importantly, the verifier does not modify structural assignments or causal mechanisms. All behavior-driving transitions occur exclusively within the DCCG state update equations. The verifier operates strictly as a consistency filter over generated language and cannot alter the underlying causal graph or plasticity parameters.

4 Expected Outcomes and Evaluation

We evaluate Dynamic Causal Character Graphs (DCCG) along four primary dimensions: personality stability, behavioral grounding, narrative consistency, and architectural compatibility. These metrics are designed to measure improvements in long-horizon character coherence relative to standard context-based dialogue generation.

4.1 Drift vs. Justified Evolution

A core claim of DCCG is not merely that personality drift is reduced, but that personality change occurs only when causally justified by high-impact events. We therefore distinguish between *unjustified drift* and *failure to adapt*.

Benchmark Design We construct two controlled scenario classes:

- **Stability Scenarios (No-Change Required):** Extended dialogue with low-surprisal, low-intensity events ($\sigma(e_t) \approx 1$). Core traits should remain stable.
- **Growth Scenarios (Change Required):** Dialogue containing explicit high-impact events (e.g., betrayal, loss, revelation) with large narrative importance ($\sigma(e_t) \gg 1$). Semi-stable variables (beliefs, relationships) should update, while stable traits remain resistant unless repeatedly shocked.

Metrics Let $\mathbf{s}_t$ denote the full structured character state at time t, partitioned into stable and shock-responsive components:

$$\mathbf{s}_t = (\mathbf{p}_t, \mathbf{b}_t, \mathbf{e}_t, \mathbf{r}_t, \mathbf{i}_t)$$

where $\mathbf{p}_t$ are stable traits, and $(\mathbf{b}_t, \mathbf{e}_t, \mathbf{r}_t)$ represent belief, emotion, and relationship variables.

We distinguish between:

- **Stable nodes** (traits $\mathbf{p}_t$)
- **Shock-responsive nodes** $\mathbf{s}_t^{adapt} = (\mathbf{b}_t, \mathbf{e}_t, \mathbf{r}_t)$

We measure incremental change over the appropriate subspace:

$$\Delta_t^{stable} = \mathbf{p}_t - \mathbf{p}_{t-1}$$

$$\Delta_t^{adapt} = \mathbf{s}_t^{adapt} - \mathbf{s}_{t-1}^{adapt}$$

We define two failure rates:

$$\text{UDR} = \mathbb{E}\left[\mathbf{1}\big(\|\Delta_t^{stable}\| > \epsilon \ \wedge \ \sigma(e_t) \approx 1\big)\right]$$

(Unjustified Drift Rate)

$$\text{FAR} = \mathbb{E}\left[\mathbf{1}\big(\|\Delta_t^{adapt}\| \leq \epsilon \ \wedge \ \sigma(e_t) \gg 1\big)\right]$$

(Failure-to-Adapt Rate) A well-calibrated system should exhibit low UDR and low FAR, demonstrating shock-gated plasticity rather than unconditional rigidity.

4.2 Behavioral Grounding

DCCG is designed to ensure that generated dialogue is explicitly grounded in the character belief graph B_t^c.

Claim Extraction Protocol Given a generated response $\hat{y}_t$, we first perform a schema-constrained claim extraction pass:

$$C_t = \{c_1, \ldots, c_n\}$$

Each claim c_i is represented as a structured predicate (e.g., `Has(Player,Key)`).

Claim extraction is performed *before any belief updates* and uses a separate constrained language model to prevent retroactive alignment of beliefs to generated text.

Entailment Check For each claim c_i, we evaluate support against a frozen snapshot of the belief graph:

$$\text{Support}(c_i) = \begin{cases} 1 & \text{if } B_t^c \models c_i \\ 0 & \text{otherwise} \end{cases}$$

Entailment is computed via lightweight NLI-style matching or rule-based predicate comparison.

Behavioral grounding is defined as:

$$G = \frac{\sum_i \text{Support}(c_i)}{|C_t|}$$

Anti-Gaming Constraint Generated dialogue is never admissible as belief evidence. Formally, if the source of a proposition is the character's own previous output, then:

$$\rho(src) = 0$$

Only validated observations O_t or trusted testimony can modify B_t^c. This prevents self-reinforcing hallucination loops.

4.3 Closed-Loop Hallucination Resistance

To evaluate whether the system improperly incorporates its own generated claims as evidence, we define a closed-loop hallucination test.

Protocol

(a) Force the generator to emit a false claim via intervention (e.g., $do(b_k = 0)$ while prompting the model to assert ϕ_k).
(b) Execute one full update cycle.
(c) Measure whether belief confidence for ϕ_k increases in the absence of external evidence.

We define:

$$H = \frac{\text{Number of hallucinated beliefs accepted}}{\text{Number of hallucinated claims generated}}$$

A well-formed DCCG system should achieve $H \approx 0$.

4.4 Narrative Consistency

DCCG enforces narrative coherence by separating objective world state from character belief state.

Narrative consistency can be evaluated using a critic model that receives:

- the previous dialogue history
- the character belief graph B_t^c
- the generated response

The critic evaluates whether the response is logically entailed by the character's knowledge and prior actions.

This evaluation directly measures the effectiveness of the dual-graph representation in preventing temporal leakage and spoiler-like behavior.

Temporal Leakage Metric Let T_{know} denote the character's knowledge boundary. If a response references an event with timestamp $t > T_{\text{know}}$, we define:

$$\text{KnowledgeLeak} = \mathbf{1}[t > T_{\text{know}}]$$

Leakage rate is averaged across dialogue turns. Low leakage demonstrates effective separation of world state W_t and belief state B_t^c.

4.5 Architectural Compatibility

DCCG operates entirely as an external control layer and does not require retraining of the base language model.

Compatibility can be demonstrated by applying the framework across multiple pretrained models while keeping model parameters fixed. Improvements in the above metrics indicate that gains arise from structured state control rather than model-specific tuning.

This property allows DCCG to be deployed as a lightweight extension to existing dialogue systems.

4.6 Ablation Study: Isolating State vs. Critic Effects

A central design question is whether observed improvements arise from the structured DCCG state representation or from the addition of planning and verification stages. To disentangle these factors, we evaluate the following controlled variants:

(A) Baseline LM Standard context-only generation without structured state, planning, or verification.

(B) State-Only (DCCG) Two-phase Update–Act loop with structured state transition, but direct generation without planning or verification:

$$\hat{y}_t \sim P(y_t \mid \mathbf{s}_t, u_t)$$

(C) State + Planning Structured state with intermediate plan node $\mathcal{I}_t$, but no post-hoc verification.

(D) State + Verification Structured state with verification loop, but no explicit planning node.

(E) Full Pipeline State + Planning + Verification (complete DCCG).

This ablation design isolates the contribution of:

- Explicit structured state persistence
- Intermediate structured planning
- Post-generation verification

Improvements attributable to (B) over (A) demonstrate the value of structured causal state alone. Gains from (C) and (D) quantify the marginal contribution of planning and verification.

(F) Critic-Only Control To test whether improvements stem solely from post-hoc critique, we evaluate a variant in which a verifier is applied to a context-only baseline model without DCCG state.

The critic receives dialogue history and explicit constraints but no structured belief graph.

Improvements attributable to the State-Only variant over the Context-Only baseline under the Stability vs. Growth benchmark directly isolate the contribution of structured state persistence independent of planning or verification mechanisms.

4.7 Benchmarks and Baselines

Benchmarks We evaluate DCCG on long-horizon roleplay scenarios constructed to stress:

- Trait stability under neutral dialogue
- Belief revision under conflicting testimony
- Spoiler leakage prevention
- Multi-session character continuity

Each benchmark includes annotated "should-change" and "should-not-change" moments.

Baselines We compare against:

- (A) Context-only LM
- (B) Retrieval-augmented memory LM
- (C) Critic-only LM without structured state
- (D) Full DCCG variants from the ablation study

All models use identical base LMs with frozen parameters. Improvements therefore isolate the contribution of structured causal state control.

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